

Lecture 5

Weds AM 7/15

outline/topics

- Action-Angle coordinates
- CS invariant + single particle
- Visualizing betatron oscillations
- Describing beam, emittance
- Adiabatic damping + normalized emittance
- Linear magnet errors + tune
- Magnetic fields from sources + higher order multipoles (intro)

Action-Angle Coordinates

H.W. 5.4.5
Lee 2.II.4

- Useful to express EOM in terms of another set of variables, action J and angle ψ (same as φ yesterday)
- Hamiltonian context:
 - ↳ transform from u, u' phase space coordinates to ψ, J coordinates
 - ↳ Canonical transformation
- Useful for describing oscillators, effects of perturbations
 - ★ review of Lagrangian + Hamiltonian mechanics in HW Chapter 4, applied to charged particle in EM field in chapter 5 (Lee Sec. 2.II.1)
- Action J is a constant of the motion, ψ is a cyclic variable

• Start from Hamiltonian of the form

$$H = \frac{1}{2} (U')^2 + \frac{1}{2} K U^2$$

Lee 2.II.4
HW 5.4.4
(general)

• Generating function: already have

$U(s)$, $\Psi(s)$ from yesterday:

$$U(s) = a \sqrt{\beta_0(s)} \cos(\Psi(s) + \xi_0)$$

$$\Psi(s) = \int_0^s \frac{ds}{\beta_0(s)} \Rightarrow \text{use } F_1(U, \Psi)$$

Find F_1 from inverting $U' = \frac{\partial F_1}{\partial U}$

$$F_1(U, \Psi) = \frac{-U^2}{2\beta} \left(\tan \Psi - \frac{\beta'}{2} \right)$$

Details on generating function form
+ derivation H.W. 5.4.5 (general)

Lee w/ β, s

New Hamiltonian

no explicit dep.
0

$$\tilde{H}(\Psi, \mathcal{J}) = H(U(\Psi, \mathcal{J}), U'(\Psi, \mathcal{J})) + \cancel{\frac{\partial F_1}{\partial s}}$$

$$J = - \frac{\partial F_1}{\partial \psi} \quad \downarrow$$

$$J = \frac{1}{2\beta} [v^2 + (\beta v' + \alpha v)^2]$$

Need $v(\psi, J)$, $v'(\psi, J)$

$$v = \sqrt{2\beta J} \cos \psi, \quad v' = -\sqrt{\frac{2J}{\beta}} (\sin \psi + \alpha \cos \psi)$$

$\alpha = -\frac{\beta}{2}$
same as 3
lec. 3
↑

⇒ Algebra! get

$$\tilde{H} = \frac{J}{\beta}, \quad \frac{dJ}{ds} = -\frac{\partial \tilde{H}}{\partial \psi} = 0$$

constant
of the motion
↳ invariant

* note other Hamilton e.g.:

$$\psi' = \frac{\partial \tilde{H}}{\partial J} = \frac{1}{\beta(s)} \quad \text{same as yesterday,}$$

• Action-angle coordinates describe an invariant phase space

* slide: (ψ, J) evolution

Courant-Snyder invariant

$$J = \frac{1}{2\beta} [u^2 + (\beta u' + \alpha u)^2]$$

· Courant-Snyder invariant

$$C(u, u') = \frac{1}{\beta} [u^2 + (\alpha u + \beta u')^2] = 2J$$

$$= \gamma_0^2 + 2\alpha u u' + \beta (u')^2 = 2J$$

This equation describes an ellipse
in u, u' coordinates

Equation of a tilted ellipse in
Cartesian coordinates:

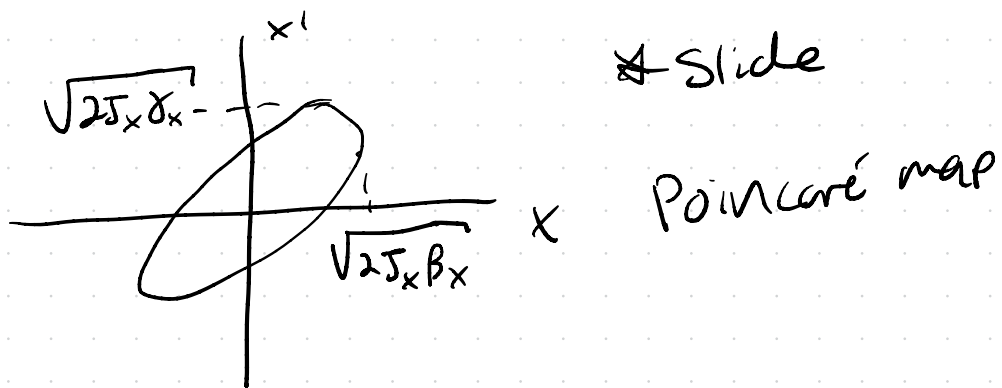
$$ax^2 + 2bxy + cy^2 = d$$

$$\text{Area of ellipse is } \frac{\pi d}{\sqrt{ac - b^2}}$$

$$\text{our case: } \gamma_0^2 + 2\alpha u u' + \beta (u')^2 = 2J$$

$$\text{area} = \frac{\pi 2J}{\sqrt{\gamma\beta - \alpha^2}} \quad \text{with} \quad \gamma = \frac{1 + \alpha^2}{\beta}$$

$$\left(\frac{1 + \alpha^2}{\beta} \right) \beta - \alpha^2 = 1 \quad \boxed{\text{area} = 2\pi J}$$



- Unaccelerated particle follows an ellipse in phase space. area enclosed is constant

↳ Keep this picture in mind, visualize betatron oscillations

Betatron oscillations / tune

• From yesterday:

$$M(s|s_0) = B(s|s_0) \begin{bmatrix} \cos \psi & \sin \psi \\ -\sin \psi & \cos \psi \end{bmatrix} B^{-1}(s|s_0)$$

• Periodic case, describe one turn of circular accelerator, M simplifies

"one-turn" Matrix

$$M = \begin{bmatrix} \cos \bar{\Phi} + \alpha \sin \bar{\Phi} & \beta \sin \bar{\Phi} \\ -\gamma \sin \bar{\Phi} & \cos \bar{\Phi} - \alpha \sin \bar{\Phi} \end{bmatrix}$$

Phase advance per turn

is also commonly used

Recall

$$\psi(s) = \int_{s_0}^s \frac{ds}{\beta(s)} \Rightarrow \bar{\Phi} = \int_s^{s+C} \frac{ds}{\beta}$$

$s+C \leftarrow$ circumference

Definition of tune:

$$Q_U = \nu_U = \frac{1}{2\pi} \oint_{s \rightarrow s+C} \frac{ds}{\beta(s)} = \frac{\Phi}{2\pi}$$

Rewrite $\Phi = 2\pi \nu_U$

$$M = \begin{bmatrix} \cos(2\pi\nu) + \alpha \sin(2\pi\nu) & \beta \sin(2\pi\nu) \\ -\gamma \sin(2\pi\nu) & \cos(2\pi\nu) - \alpha \sin(2\pi\nu) \end{bmatrix}$$

Evolution over one turn depends
on the tune ~~slide~~

- So far, focused on trajectories of a single particle undergoing betatron oscillations around ideal closed orbit

- Now $\Rightarrow$ beam as a whole

Emittance

- Describe beam by describing envelope
- Real beam distributed in x, x' and z, z' coordinates
- Total phase space area of the beam is conserved H.W. 8.1.1
 - ↳ Liouville's Theorem Lee A.I.4
- Beam: collection of particles $\Rightarrow$ identical dynamical systems governed by same Hamiltonian, varying initial conditions X slide
- Beam case: each particle has its CS ellipse

↳ can describe emittance $C(u, u') = \epsilon$

$$\Rightarrow \text{Area enclosed} = 2\pi J$$

$$C(u, u') = 2J \Rightarrow \text{Area} = \pi \epsilon$$

units of $\pi [\text{length}] \cdot [\text{angle}]$

$$\xi = \frac{\text{Area}}{\pi} \text{ units of length} \cdot \text{angle} \quad \text{ie. mm-mrad}$$

(although sometimes π is absorbed into the angle)

- Different definitions of description of beam as a whole

↳ envelope, describing that size
 ↳ varying definitions/uses capturing different fraction of particles

- Distribution of particles, Lee 2.II.5.A describe widths of distribution

$$\begin{array}{ccc} \sigma_U, \sigma_{U'}, \sigma_{UU'} \\ \swarrow \quad \downarrow \quad \searrow \\ \text{in } U \quad \text{in } U' \quad \text{correlation} \end{array}$$

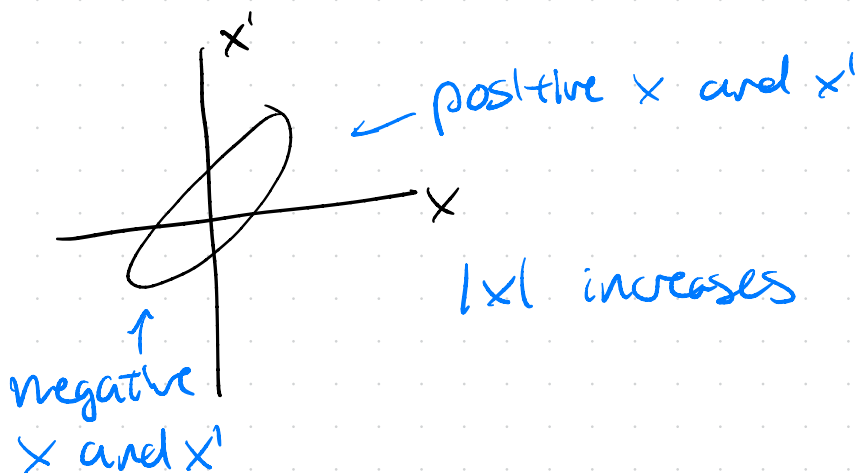
$$\text{e.g. } \xi_{\text{rms}} \equiv \sqrt{\sigma_U^2 \sigma_{U'}^2 - \sigma_{UU'}^2}$$

* see uploaded extra notes

- "Total boundary" enclosing 95% of all particles, or 99% is also common

- Evolution through linear elements
(our elements so far)

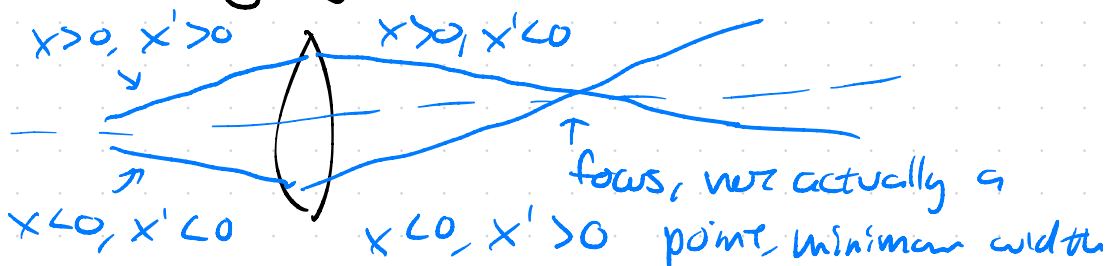
Rotates envelope of beam $\rightarrow$ slides



$\rightarrow$ recall our drift

$$M = \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{aligned} x(s) &= x_0 + x'_0 l \\ x'(s) &= x'_0 \end{aligned}$$

converging + diverging beam:



- Recall: propagate $\beta(s), \alpha(s), \gamma(s)$
↳ Lec 3

- Propagate β, α, γ for this ellipse, i.e.
through drift space L , like pset 2
↳ refer to Lec 2.32
H.W. example in 8.1.2

$$\alpha(s) = -\frac{\beta'}{2} \quad \text{if at min/max in } \beta, \\ \alpha = 0$$

- Other example slides

Adiabatic damping

- $C(u, u')$ is not invariant with changing energy, $p \neq p_0$

↳ decreases with increased momentum

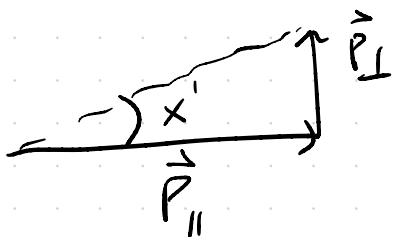
- So far, neglecting acceleration, but if we consider changing p , simple geometrical picture shows how

$$x' = \frac{dx}{ds} = \frac{p_{\perp}}{p_{\parallel}}$$

SY Lee 2II.5.E

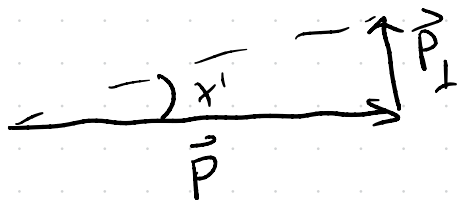
$$p_0 = p_{0'} \\ = mc\beta\gamma u'$$

H.W. 10.3.2



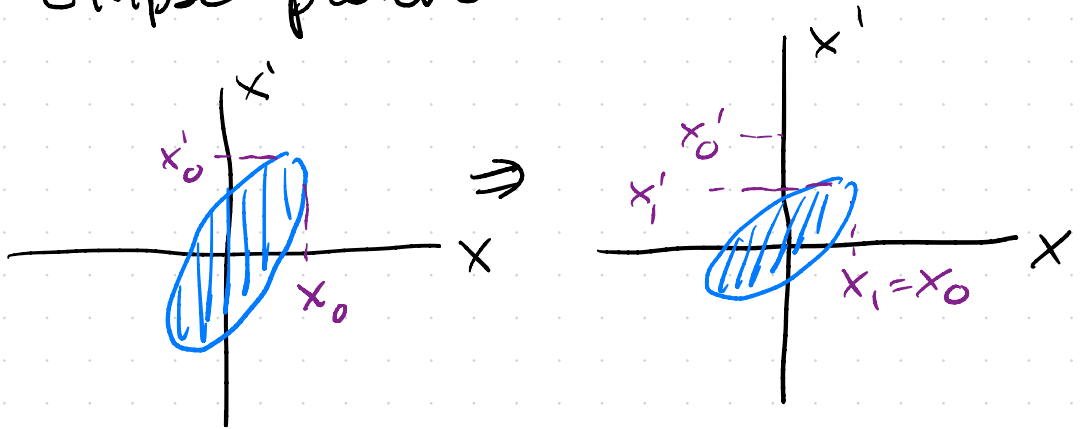
acceleration increases

$\dot{p}_{\parallel}$ not $\dot{p}_{\perp}$



x' decreases

Ellipse picture:



- x extent does not change, but x' decreases, reduced area
- Define normalized emittance

$$\Sigma_N = \beta \gamma \epsilon$$

↓
invariant over
acceleration

these are relativistic
 β, γ

* does not account for synchrotron radiation, damping, dispersive effects

Linear magnet errors

Back to Lecture 2, had for error-free linear magnets:

$$x'' + k_x(s)x = 0 \quad z'' + k_z(s)z = 0$$

• Add some error term

Lee 2.III

$$\Rightarrow x'' + k_x(s)x = \frac{\Delta B_z}{B\rho}$$

$$z'' + k_z(s)z = -\frac{\Delta B_x}{B\rho}$$

ΔB_z and ΔB_x are perturbations

equations are now inhomogeneous

- Two types today: dipole field errors (errors in bending) + quad field errors (errors in focusing)

- Error in bending field ΔB
 from ideal over some path length l
 $\Rightarrow$ delivers kick $\theta = \frac{\Delta B l}{(B\rho)}$ *still rigidity*

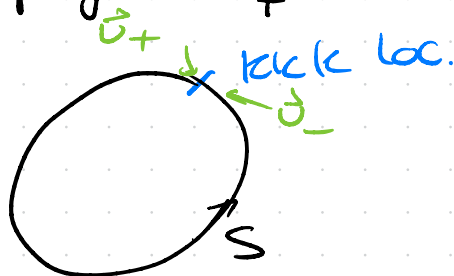
- Impacts our closed orbit!
 $\hookrightarrow$ New perturbed closed orbit

if $\vec{U}_+ = \begin{bmatrix} U_0 \\ U'_0 \end{bmatrix}$ immediately after kick,

$\vec{U}_- = \begin{bmatrix} U_0 \\ U'_0 - \theta \end{bmatrix}$ immediately before kick

Closed orbit condition, one turn
 matrix needs to propagate $\vec{U}_+$
 back to $\vec{U}_-$

$$\vec{U}_- = M \vec{U}_+$$



$$M \begin{bmatrix} u_0 \\ u'_0 \end{bmatrix} = \begin{bmatrix} u_0 \\ u'_0 - \theta \end{bmatrix}$$

Can also write as

$$M \begin{bmatrix} u_0 \\ u'_0 \end{bmatrix} = \begin{bmatrix} u_0 \\ u'_0 \end{bmatrix} - \begin{bmatrix} 0 \\ \theta \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0 \\ \theta \end{bmatrix} = \begin{bmatrix} u_0 \\ u'_0 \end{bmatrix} - M \begin{bmatrix} u_0 \\ u'_0 \end{bmatrix} = (I - M) \begin{bmatrix} u_0 \\ u'_0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} u \\ u'_0 \end{bmatrix} = (I - M)^{-1} \begin{bmatrix} 0 \\ \theta \end{bmatrix}$$

* Remember what is M ?

for one turn:

$$M = \begin{bmatrix} \cos \Phi + \alpha \sin \Phi & \beta \sin \Phi \\ -\gamma \sin \Phi & \cos \Phi - \alpha \sin \Phi \end{bmatrix}$$

Rewrite $\Phi = 2\pi\nu U$ (same as earlier)

$$M = \begin{bmatrix} \cos(2\pi\nu) + \alpha \sin(2\pi\nu) & \beta \sin(2\pi\nu) \\ -\gamma \sin(2\pi\nu) & \cos(2\pi\nu) - \alpha \sin(2\pi\nu) \end{bmatrix}$$

↳ solve

$$\Rightarrow U_0 = \frac{\beta \cos(\pi\nu_0) \Theta}{2 \sin(\pi\nu_0)}$$

$$U'_0 = \frac{\Theta}{2 \sin(\pi\nu_0)} [\sin(\pi\nu_0) - \alpha_0 \cos(\pi\nu_0)]$$

at Kretz location

At any other s : (propagate $\vec{U}_0$)

$$U(s) = \frac{\Theta \sqrt{\beta(s)} \beta_0}{2 \sin \pi\nu_0} \cos(\psi(s) - \pi\nu)$$

integer tune $\nearrow$ periodic closed ~~slide~~
orbit not possible $\Rightarrow$ Kretz add

• Much more detail in remaining Lee 2.III.1 sections

• Lee 2.III.3 discusses applications of intentional dipole "error"

↳ e.g. kick beam from stable orbit for extraction

• Extended transfer matrix description is also useful to introduce:

$$\begin{bmatrix} U \\ U' \\ 1 \end{bmatrix}_{S=S_2} = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} m_{13} \\ m_{23} \\ 1 \end{bmatrix} \begin{bmatrix} U \\ U' \\ 1 \end{bmatrix}_{S=S_1}$$

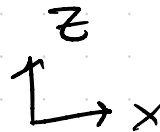
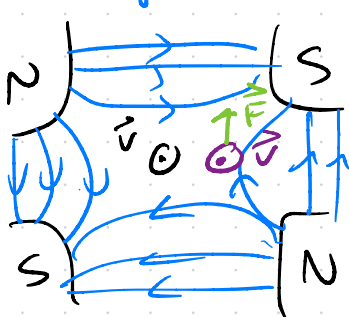
perturbation terms

original M

SY Lee 2.III.2

↳ example for combined quad-dipole

↳ misalignment of quad $\Rightarrow$ focusing error



shift $\updownarrow$ or $\leftrightarrow$

Quadrupole field (gradient) errors

- Error in gradient: e.g. for motion around closed orbit

Lee 2 III.4

$$v'' + [K_0(s) + K(s)]v = 0$$

$K(s)$ for ideal lattice

error $K(s)$ in

- sees error each turn:

$$K(s) = K_0(s) + K(s) = K_s(s + C)$$

circumference

- Unperturbed one-turn transfer matrix

$$M_0 = I \cos \bar{\Phi}_0 + J \sin \bar{\Phi}_0$$

$\bar{\Phi}_0 = 2\pi \nu_0$ unperturbed

J ideal phase advance

$$J = \begin{bmatrix} \alpha(s) & \beta(s) \\ -\gamma(s) & -\alpha(s) \end{bmatrix} \text{ unperturbed}$$

localized gradient error: $ds_1 = \text{size}$

$$m(s_1) = \begin{bmatrix} 1 & 0 \\ -K(s_1)ds_1 & 1 \end{bmatrix}$$

↑ strength · length

Perturbed one-turn matrix:

$$M(s_1) = M_0(s_1)m(s_1)$$

Analogous to previous discussion,
same perturbed phase advance Φ

$$\Delta\Phi = \Phi - \Phi_0 \quad \text{phase shift}$$

$$\Delta\Phi \approx \frac{1}{2} \beta_1 K(s_1) ds_1$$

⇒ tune shift

$$\Delta\nu = \frac{1}{4\pi} \beta_1 K(s_1) ds_1 \quad \text{localized}$$

↑ unperturbed β at s_1

- Also modulates β betatron amplitude
 ↳ essentially think of it as beating

↳ Lee 2.III.1.B

HW 15.3.4

↳ result:

$$\frac{\Delta \beta}{\beta} = - \frac{1}{2 \sin \Phi_0} \int_{s_0}^{s_0 + L} K(s) \beta(s) \cos[2\nu_0(\pi + \psi - \psi_0)] ds,$$

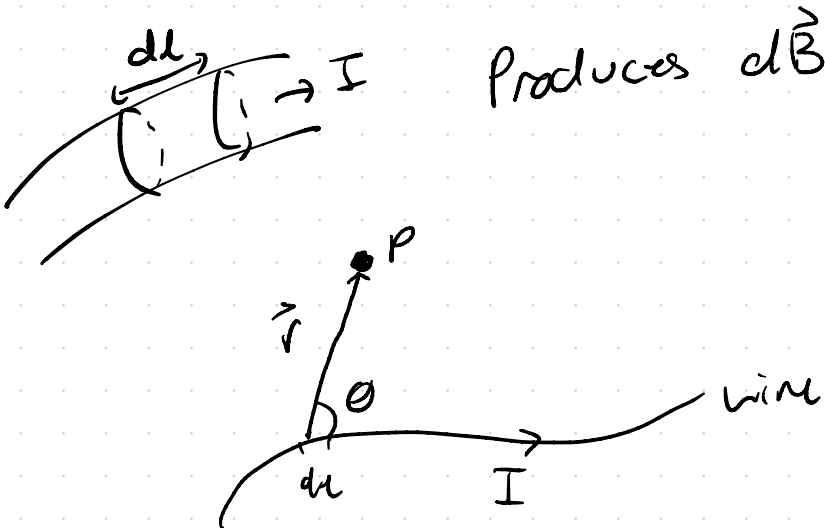
$$2 \sin \Phi_0 = 2 \sin(2\pi \nu_0)$$

Blows up for half-integer tunes!

⇒ Both integer + half-integer should
 be avoided
 + others! slide

Magnetic fields

- Two types of magnets: electromagnets and permanent magnets
- Electromagnets have benefit of being variable
 - ↳ "ramp" magnets for different energies
- Field produced by a current
- Find field? Biot-Savart Law:



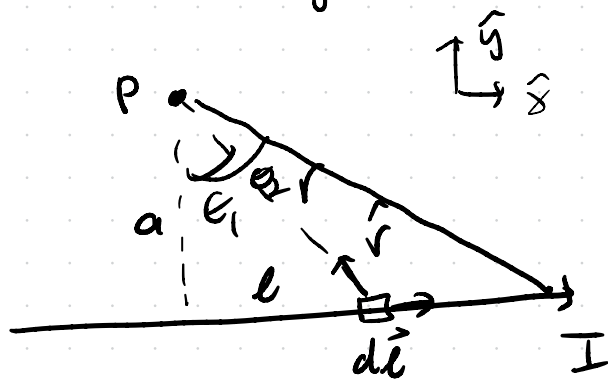
$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \vec{r}}{r^2} \quad \vec{r} = r \hat{r}$$

Find distribution at any point in space,
arbitrary wire

← steady current case

$$\vec{B} = \int_{\text{wire}} d\vec{B} = \frac{\mu_0 I}{4\pi r^2} \int_{\text{wire}} d\vec{l} \times \hat{r}$$

Finite straight wire:



$d\vec{B}$ out of page at P from all elements
(RHR)

$$\cos \theta = \frac{a}{r} \quad \sin \theta = \frac{l}{r}$$

$$d\vec{l} = dl \hat{x}$$

$$\hat{r} = -\sin \theta \hat{x} + \cos \theta \hat{y}$$

$$d\vec{l} \times \hat{r} = dl \cos \theta \hat{z} \text{ at } P$$

$$\cos \theta = \frac{a}{r} \quad \sin \theta = \frac{l}{r}$$

$$d\vec{l} = dl \hat{x}$$

$$\hat{r} = -\sin \theta \hat{x} + \cos \theta \hat{y}$$

$$d\vec{l} \times \hat{r} = dl \cos \theta \hat{z} \text{ at } P$$

Using magnitude since we know direction

$$\tan \theta = \frac{l}{a} \quad \text{so} \quad l = a \tan \theta$$

$$dl = a \sec^2 \theta d\theta = \frac{a}{\cos^2 \theta} d\theta$$

$$d\vec{l} \times \hat{r} = dl \cos \theta = \left(\frac{a}{\cos^2 \theta} d\theta \right) \cos \theta$$

$$r = \frac{a}{\cos \theta} \Rightarrow \frac{1}{r^2} = \frac{\cos^2 \theta}{a^2}$$

$$dB = \frac{\mu_0 I}{4\pi r^2} \frac{d\vec{l} \times \hat{r}}{r^2} = \frac{\mu_0 I}{4\pi} \left(\frac{a}{\cos^2 \theta} d\theta \right) \cos \theta \left(\frac{\cos^2 \theta}{a^2} \right)$$

$$B = \frac{\mu_0 I}{4\pi a} \int_{\theta_1}^{\theta_2} \cos \theta d\theta = \frac{\mu_0 I}{4\pi a} \sin \theta \Big|_{\theta_1}^{\theta_2}$$

$$\vec{B}(P) = \frac{\mu_0 I}{4\pi a} (\sin \theta_2 - \sin \theta_1) \hat{z}$$

• For finite wire: same θ_1 and θ_2

• For long/infinite wire: $\theta_1 = -\frac{\pi}{2}$, $\theta_2 = \frac{\pi}{2}$

$$\vec{B}(P) = \frac{\mu_0 I}{4\pi a} \left(\sin\left(\frac{\pi}{2}\right) - \sin\left(-\frac{\pi}{2}\right) \right) \hat{z}$$

$$\vec{B} = \frac{\mu_0 I}{2\pi a} \hat{z} \quad \text{at } P$$

Any radius r : $B(r) = \frac{\mu_0 I}{2\pi r}$

Direction: RHR or:

$$\hat{z} \times \hat{r} = \hat{\phi} \text{ cycl.}^*$$

Multipole expansion

• Complex representation of field

H.W. 6.1
Use 2.III into

$$B_y(x, y) + i B_x(x, y)$$

- can be represented by complex power series,

$$B_y + iB_x = \sum_{n=0} C_n (x+iy)^n$$

$\Rightarrow$ multiple expansion representation
 n is order of multiple

$$C_n = b_n + i a_n \text{ or } \lambda_n + i \mu_n$$

(most places.) H.W.

b_n or λ_n : upright, a_n or μ_n : skew
~~slides~~

$$B_y + iB_x = B_{ref} \sum_{n=1}^{\infty} (b_n + i a_n) (x+iy)^{n-1}$$

here $n=1$ dipole

Lee uses

$$\sum_{n=0}^{\infty} (b_n + i a_n) (x+iy)^n$$

$n=0$ dipole